MEDICATION SAFETY ASSISTANT

Software Development Plan

(Small Project)

Version 1.0

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1. Introduction

1.1 Purpose

This Software Development Plan defines how the Medication Safety Assistant project will be organized, executed, monitored, and controlled. It provides a shared working agreement for the project team and gives the instructors and teaching assistants a basis for reviewing the team's plan and progress. The document covers the initial lifecycle plan, team responsibilities, iteration objectives, planned releases, requirements management, and risk management.

This is the initial plan produced during the Inception phase. It will be reviewed at the end of each sprint and updated when scope, risks, estimates, or course milestones change.

1.2 Scope

This plan applies to the design and development of the Medication Safety Assistant, a responsive Web application that helps patients understand potential medication-related risks and follow confirmed medication schedules. The planned product accepts printed or digital Vietnamese prescriptions, manual medication entries, typed requests, and voice input. OCR converts prescription images into editable text, speech-to-text converts recorded speech into an editable transcript, and NLP identifies the user's intent and relevant medication information. It supports user verification, medication normalization, drug–drug interaction assessment, safety guidance, professional review of high-risk or uncertain cases, reminders, adherence tracking, and recording of possible adverse drug reactions.

The plan covers the three RUP phases required for the course—Inception, Elaboration, and Construction—implemented through five two-week sprints. It does not cover production deployment to a healthcare organization, integration with real clinical systems, or operation as a regulated clinical decision system.

1.3 Overview

Section 2 describes the project purpose, constraints, and deliverables. Section 3 defines the team structure and responsibilities. Section 4 describes the lifecycle, sprint objectives, planned builds, schedule, resources, requirements management, and risk management. Project estimates, reporting and measurement, and configuration management are intentionally left blank in this PA1 version as directed by the assignment brief.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

Many patients, particularly people with chronic conditions or prescriptions from several healthcare professionals, have difficulty determining whether their medications are safe to use together. Risks include drug–drug interactions, duplicate active ingredients, contraindications, missed doses, and possible adverse drug reactions. Prescription readability and different brand names for related active ingredients make the problem harder.

The project will produce a Medication Safety Assistant that helps users answer the question: "What potential safety risks should I be aware of in my current medications, and when should I seek professional advice?" The product supports prescription-image OCR, manual entry, typed requests, and voice input. Speech-to-text transcribes a user's voice, while NLP identifies relevant intents and medication information from the resulting text. The system requires the user to verify all OCR-extracted or voice-transcribed details before using them in an assessment. It first identifies Vietnamese products and active ingredients against the medicine-registration data published by the Drug Administration of Vietnam (DAV), then maps the normalized active ingredients to DDInter 2.0 entities for international drug–drug interaction and severity checking. AI may explain structured system findings, but it must not independently determine medication safety.

The project objectives are:

- Provide multimodal medication and request input through prescription OCR, manual entry, typed conversation, and voice-to-text, with an editable confirmation step before further processing;

- Apply NLP to verified text to identify user intent and relevant medication information without allowing the language model to make independent medication-safety conclusions;
- Identify Vietnamese medicine names, active ingredients, strengths, and registration information using data published by the Drug Administration of Vietnam;
- Map the identified active ingredients to corresponding DDInter 2.0 entities and assess supported drug–drug interaction risks and their severity using international interaction data;
- Clearly distinguish detected risks, no known interaction within the available data, and incomplete or unassessed cases;
- Provide understandable safety guidance and route high-risk or uncertain cases to professional review;
- Create reminders only from confirmed prescription schedules and record adherence and possible adverse drug reactions; and
- Deliver a demonstrable Web prototype within the five-sprint course schedule.

The initial release excludes advanced timing analysis, drug–food interaction checks, pill recognition, handwritten prescription support, and integration with real healthcare services.

2.2 Assumptions and Constraints

Type	Description	Effect on the project
Assumption	Users can review and correct OCR results and voice transcripts before assessment.	Safety checks run only on user-verified medication details, never directly on an unconfirmed OCR result or speech transcript.
Assumption	The medicine-registration data published by the Drug Administration of Vietnam (DAV) is available as the primary source for identifying Vietnamese medicinal products and their active ingredients.	The team must preserve source provenance and account for corrections, changes, supplements, or withdrawals that may not be reflected in the original registration publication.
Assumption	Vietnamese active ingredients can be mapped reliably to corresponding DDInter 2.0 entities for interaction checking.	Mapping rules, ambiguous matches, coverage, and permitted DDInter access must be validated early; unresolved mappings remain unassessed rather than safe.
Assumption	Professional review is simulated for the course prototype.	The prototype demonstrates the workflow but does not connect to real clinicians or clinical systems.
Assumption	A responsive Web application is sufficient for desktop and mobile users.	The team can maintain one codebase and test on representative modern browsers.
Schedule constraint	The course requires three phases and five sprints with a small three-person team.	Scope must be prioritized around one end-to-end medication safety workflow.
Safety constraint	The system must not prescribe, stop, replace, or modify prescription medication.	All messages must use cautious language and direct users to qualified professionals when appropriate.
Data constraint	Missing data or no detected interaction is not proof of safety.	The interface must display coverage limitations and an incomplete/unassessed state.

Reminder constraint	New reminders for medication involved in a severe interaction require professional review and confirmation.	A safety gate is required before reminder activation; unrelated medication reminders may proceed.
Privacy constraint	Prescription images, medication lists, health information, and symptoms are sensitive.	The prototype must minimize collected data, restrict access, avoid exposing secrets, and use non-identifying test data where possible.
Technical constraint	Initial OCR supports printed or digital Vietnamese prescriptions, not handwriting.	Low-confidence and unsupported input must be corrected manually or reported as unresolved.
Technical constraint	Voice-to-text accuracy varies with accent, pronunciation, background noise, microphone quality, and medication terminology.	The interface must display an editable transcript and require confirmation; low-confidence or unresolved medication terms must not be assessed automatically.
Privacy constraint	Voice recordings and transcripts may contain sensitive health information.	Audio collection requires clear permission, minimal retention, restricted access, and deletion after transcription when storage is not necessary.

2.3 Project Deliverables

Deliverable	**Description / acceptance condition**	**Owner**	**Target**
PA1 planning package	Initial Project Plan, Vision Document, 3–6 UI screens, two weekly reports, and accessible tracking evidence.	Whole team	Sprint 1
Requirements baseline	Reviewed vision, prioritized backlog, multimodal-input use cases, non-functional requirements, and traceability to interview needs.	Long	Sprint 2
Architecture and UX baseline	System context, main workflow, data/interface decisions, and tested UI prototype covering OCR, manual, typed, and voice input.	Trang / Thành	Sprint 2
Multimodal intake proof of concept	OCR extraction, speech-to-text transcription, NLP intent/medication processing, editable confirmation, and consistent verified-text output.	Long / Trang / Thành	Sprint 3
Risk-checking proof of concept	Prepared DAV medicine data, reviewed active-ingredient mappings to DDInter, and a controlled drug–drug interaction query with explicit result states.	Long	Sprint 3
Integrated test build	End-to-end Web build covering multimodal medication intake, verification, normalization, safety results, and safety guidance.	Whole team	Sprint 4
Final prototype release	Stable course prototype including voice input, confirmed-schedule reminders, basic adherence/ADR recording, regression results, documentation, and demo material.	Whole team	Sprint 5

Project evidence repository	Versioned source, decisions, test evidence, sprint records, risk register, meeting notes, and weekly reports accessible to teaching staff.	Trang	Continuous

3. Project Organization

3.1 Organizational Structure

The project is executed by a self-managing team of three students. Hoàng Yến Trang is the Team Leader and Project Manager and is the primary contact with the instructors and teaching assistants. Planning decisions are discussed by the whole team. A decision that changes product scope, medical-safety behavior, or a committed sprint objective requires team review and must be recorded in the project workspace.

Work is organized as a shared product backlog and five sprint backlogs. Each work item has one accountable owner, acceptance criteria, and a tracked status. Another member reviews safety-critical requirements, code, and test evidence before the item is accepted. The proposed role allocation below is an initial working arrangement and may be adjusted at sprint retrospectives according to workload and demonstrated skills.

3.2 Roles and Responsibilities

Role	Assigned member(s)	Responsibilities
Project Manager / Team Leader	Hoàng Yến Trang	Maintain the plan and sprint goals; coordinate meetings; track dependencies, risks, and blockers; communicate with teaching staff; ensure evidence and deliverables are submitted on time.
NLP and Conversational Processing Developer	Cao Hoàng Long	Implement intent identification and medication-entity processing for typed and transcribed requests; constrain generated explanations to structured safety findings; test multi-intent, ambiguous, and unsafe requests; maintain NLP acceptance criteria.
OCR and Prescription Intake Developer	Hoàng Yến Trang	Implement image upload/capture, preprocessing, OCR extraction, confidence/error handling, and the editable prescription-verification flow; test printed and digital Vietnamese prescriptions and unsupported handwriting cases.
Voice Input and Speech-to-Text Developer	Lỗ Chí Thành	Implement microphone permission, audio capture, speech-to-text integration, medication-term transcription, confidence/error handling, and the editable transcript-confirmation flow; minimize retention of sensitive audio.
Medication Data Preparation and Risk Integration Developer	Cao Hoàng Long	Collect, inspect, clean, deduplicate, and version the DAV medicine-registration data; normalize Vietnamese products and active ingredients; prepare and review mappings to DDInter 2.0; create controlled test datasets; maintain data provenance, coverage notes, and explicit assessment states.
UI/UX and Accessibility Reviewer	Hoàng Yến Trang	Design and evaluate responsive multimodal input, verification, result, warning, reminder, and follow-up interfaces; ensure that voice input has an accessible manual fallback.

Safety and Quality Reviewers	All members	Review safety-critical behavior; design positive, negative, boundary, and incomplete-data tests; verify that severe or uncertain cases are not presented as safe; record defects and regression evidence.
Configuration and Build Coordinator	Hoàng Yến Trang	Maintain repository structure, branch and pull-request conventions, development configuration, reproducible builds, tagged releases, and recovery copies of project artifacts.
Documentation and Evidence Coordinator	Hoàng Yến Trang; Cao Hoàng Long	Maintain reports, diagrams, meeting notes, UI evidence, accessible document links, and consistent terminology across the proposal, plan, vision, backlog, and prototype. Trang coordinates final document and evidence reviews as Team Leader.

4. Management Process

4.1 Project Estimates

4.2 Project Plan

The project follows an iterative, risk-driven process. Each sprint begins with planning and ends with a review and retrospective. Safety-critical assumptions and the highest technical risks are addressed before secondary features. The dates below form an initial working schedule anchored to the PA1 deadline of 18 October 2026. They must be aligned with the official course calendar if the teaching staff announces different milestones.

4.2.1 Phase and Iteration Plan

Phase	Sprint	Dates	Objective	Milestone / exit criteria
Inception	Sprint 1	5–18 Oct 2026	Agree on the problem, users, scope, team process, major risks, initial plan, and product vision.	PA1 package completed; tracking tools active; 3–6 main UI screens and two weekly reports available.
Elaboration	Sprint 2	19 Oct–1 Nov 2026	Baseline the critical requirements, architecture, data strategy, safety states, and principal user journey.	Reviewed backlog and use cases; architecture/UX baseline; data-source feasibility recorded.
Elaboration	Sprint 3	2–15 Nov 2026	Retire the highest OCR, speech-to-text, NLP, normalization, interaction-data, and safety-communication risks through an executable proof of concept.	OCR, voice, and typed input converge on verified text; medication identity and a controlled interaction check are demonstrated.
Construction	Sprint 4	16–29 Nov 2026	Implement and integrate the primary end-to-end medication safety workflow.	Test build supports intake through guidance; critical path passes integration tests.

Construction	Sprint 5	30 Nov–13 Dec 2026	Complete prioritized reminder/follow-up capabilities, stabilize the prototype, and prepare the final demonstration.	Final prototype and documentation accepted; critical defects closed; demo and evidence ready.

Iteration objectives.

Sprint	Objectives	Key work items	Owner(s)
Sprint 1	Validate the project direction and establish a controllable project.	Refine proposal; draft Project Plan and Vision; create UI sketches; set up repository and tracker; define backlog, risks, working agreements, and weekly reports.	Long, Trang, Thành
Sprint 2	Define what will be built and how the safety-critical workflow will work.	Analyze interview needs; write use cases and NFRs; prototype OCR, manual, typed, and voice flows; define NLP, OCR, speech-to-text, data, privacy, and safety interfaces and acceptance tests.	Long, Trang, Thành
Sprint 3	Prove the highest-risk technical and safety assumptions.	Long implements NLP intent/entity processing and prepares, cleans, versions, and maps DAV/DDInter data; Trang implements OCR and image-based prescription intake; Thành implements voice capture and speech-to-text. Integrate all modes through editable verification and explicit assessment states.	Long, Trang, Thành
Sprint 4	Deliver an integrated core product.	Implement primary pages and APIs; persist verified medication data; run automatic checks; show severity and limitations; simulate professional review; conduct system and usability testing.	Whole team
Sprint 5	Complete, stabilize, and demonstrate the prioritized release.	Add confirmed-schedule reminders, severe-interaction gate, adherence and possible-ADR records; fix defects; run regression and acceptance tests; finalize documents and demo.	Whole team

4.2.2 Builds and Releases

Four executable builds are planned. Additional internal builds may be produced by continuous integration, but the following are the review and test baselines.

Build / release	Type	Target date	Purpose and included scope
UI Prototype	Design prototype	18 Oct 2026	Validate navigation and OCR, manual, typed, and voice input; include transcript/extraction confirmation, results, warnings, and reminders; no clinical conclusions are produced.

Build 0.1	Technical proof of concept	15 Nov 2026	Test OCR, voice-to-text, NLP, manual input, editable confirmation, DAV-based product identification, DDInter mapping, controlled interaction lookup, and explicit assessment states.
Build 0.2	Integrated test build	29 Nov 2026	Provide the end-to-end core workflow for team, usability, integration, and safety testing.
Build 1.0	Final course prototype	13 Dec 2026	Add prioritized reminder and follow-up functions, resolve critical defects, and provide a stable demonstration baseline.

Each baseline is tagged in version control and accompanied by a short release note containing included scope, known limitations, test status, and unresolved risks. A build is not described as safe for clinical use.

4.2.3 Project Schedule

The detailed schedule is maintained in the team's task-tracking tool. The following table is the initial high-level baseline; individual backlog items will be assigned and updated during sprint planning.

Period	**Workstream**	**Key scheduled work**	**Accountable owner**	**Output**
5–11 Oct	Product discovery	Consolidate proposal, problem statement, users, features, constraints, and initial risks.	Long	Scope baseline
12–18 Oct	PA1 planning	Complete plan/vision, UI prototype, tracker setup, weekly reports, and submission review.	Whole team	PA1 package
19–25 Oct	Requirements and UX	Interview synthesis; OCR, voice, typed, and manual-input use cases; editable verification; acceptance criteria; terminology.	Long / Trang / Thành	Requirements draft
26 Oct–1 Nov	Architecture and data preparation	Long prepares the DAV dataset, active-ingredient normalization rules, DDInter mapping approach, provenance, and coverage notes; the whole team defines multimodal interfaces, privacy boundaries, and the data model.	Long (data); whole team (architecture)	Baseline package
2–8 Nov	Multimodal intake spike	NLP and intent processing (Long); OCR and prescription extraction (Trang); voice capture and speech-to-text (Thành); shared verification contract.	Long / Trang / Thành	Working input slices

9–15 Nov	Intake and assessment integration	Long integrates the prepared DAV dataset and reviewed DDInter mappings; the team connects verified text from all input modes and tests severity, unresolved mappings, incomplete states, and explanation constraints.	Long leads; Trang and Thành support	Build 0.1
16–22 Nov	Core implementation	Implement and integrate intake, medication list, automatic checks, results, warnings, and limitations.	Whole team	Integrated alpha
23–29 Nov	Core verification	Professional-review simulation, system tests, usability checks, defect correction, release notes.	Whole team	Build 0.2
30 Nov–6 Dec	Follow-up features	Confirmed-schedule reminders, safety gate, adherence and possible-ADR records.	Trang / Thành	Feature-complete beta
7–13 Dec	Stabilization and handoff	Regression, acceptance review, critical defect closure, documentation, demo rehearsal, final baseline.	Whole team	Build 1.0

Dependencies are managed in this order: OCR extraction, speech transcription, manual entry, and typed conversation each produce editable text; NLP processes the applicable text; user verification precedes identification against DAV data; Vietnamese product identification precedes active-ingredient mapping to DDInter 2.0; successful mapping precedes interaction assessment; and assessment status and severity precede guidance and reminder activation. Documentation, risk review, testing, and tracking continue throughout all sprints.

4.2.4 Project Resourcing

Phase / sprint	**People / roles**	**Skills and tools**	**Training or resource need**
Inception / Sprint 1	Three team members; team leader coordinates.	LaTeX/Docs, Figma or equivalent, Jira or equivalent tracker, Git repository.	Agree on workflow, definition of done, medical-safety language, and document access.
Elaboration / Sprint 2	Long leads NLP and medication-data preparation; Trang leads OCR; Thành leads speech-to-text; all review requirements and safety.	NLP evaluation, OCR evaluation, Web audio/microphone flows, responsive UX, architecture, DAV data preparation, and DDInter mapping analysis.	Long studies source fields, provenance, coverage, and terminology; the team defines consent, audio privacy, the verified-text schema, and representative scenarios.
Elaboration / Sprint 3	Long owns NLP and all prepared DAV/DDInter data artifacts; Trang owns OCR; Thành owns voice-to-text; all integrate and test.	Language/intent tests, data cleaning and versioning, active-ingredient normalization and mapping, image preprocessing/OCR, speech recognition, and automated integration tests.	Long creates controlled medication and mapping datasets; Trang and Thành provide representative image and audio inputs for integration tests.

Construction / Sprint 4	All developers and testers.	Web implementation, integration, browser testing, defect tracking, peer review.	Cross-train members on the integrated flow and failure-state testing.
Construction / Sprint 5	All members; leader coordinates release.	Reminder logic, regression/acceptance testing, release management, presentation.	Long prepares the fallback demonstration dataset; the whole team verifies submission links and artifacts.

The project will use student laptops, modern desktop and mobile browsers, a shared task tracker, a Git-based repository, a shared document workspace, and approved external APIs or controlled datasets. No real patient data is needed for development or demonstration.

4.3 Project Monitoring and Control

4.3.1 Requirements Management

The Vision document is the source of high-level stakeholder needs and product features. Detailed requirements are represented as backlog items and use cases in the task tracker. Each requirement or story receives a stable identifier, priority, accountable owner, acceptance criteria, and links to related design, implementation, and test evidence. Safety-critical items are labeled and must include negative and incomplete-data acceptance criteria.

A proposed requirement change is recorded as a change item rather than applied informally. The team analyzes its user value, safety impact, dependencies, effort, schedule effect, and effect on committed sprint work. The Project Manager and the member responsible for the affected area review the proposal; changes to scope or safety behavior require agreement of the whole team. An accepted change updates the Vision document or applicable specification, the backlog and priority, acceptance tests, and traceability links. Rejected or deferred changes remain recorded with their rationale.

Requirements are reviewed during sprint planning, clarified during the sprint, demonstrated at the sprint review, and reconsidered during the retrospective. No requirement is considered complete until its acceptance criteria and linked tests pass and its user-facing limitations are documented.

4.3.2 Reporting and Measurement

4.3.3 Risk Management

Risks are identified from interviews, requirements, architecture work, external dependencies, implementation, and testing. The team assesses each risk by likelihood and impact and assigns an owner. The risk register is reviewed at least during weekly coordination and at each sprint planning and review meeting. A risk is closed only when its trigger can no longer occur or the mitigation has been verified; realized risks become tracked issues.

ID	Ranking	Description and impact	Mitigation / contingency	Owner	Status
R-01	High	A Vietnamese product or active ingredient may be identified incorrectly from DAV data or mapped to the wrong DDInter 2.0 entity, producing a misleading interaction result.	Require user verification; retain DAV source identifiers and mapping provenance; use reviewed mapping rules; test aliases and ambiguous ingredients; keep unresolved mappings unassessed.	Long	Open

R-02	High	OCR errors could change a medication name, strength, or schedule and lead to an invalid assessment.	Display extracted fields for confirmation; show confidence/validation errors; provide manual correction; never assess unverified extraction.	Trang	Open
R-03	High	AI-generated explanations could add unsupported medical conclusions or false reassurance.	Generate explanations only from structured findings; constrain prompts/output; prohibit independent safety conclusions; add disclaimers and professional escalation; test unsafe prompts.	Long	Open
R-04	High	Speech-to-text may confuse Vietnamese accents, noisy speech, or medication terminology, causing an incorrect medication or schedule.	Show an editable transcript; require confirmation; highlight low-confidence terms; offer manual input; test accents, noise, dosages, and similar medication names; never assess unverified speech.	Thành	Open
R-05	High	DAV publication limitations, DDInter coverage/access, or incompatibility between the two data sources may prevent reliable mapping and assessment.	Verify both sources in Elaboration; document dates and coverage; store only permitted data; maintain a reviewed mapping table; use a controlled demonstrator dataset or narrow scope if needed.	Long	Open
R-06	High	Users may interpret “no known interaction found” as a guarantee of safety.	Use distinct result states and cautious wording; always display data limits; usability-test comprehension; route uncertainty for review.	Trang / Long	Open
R-07	Medium	Sensitive prescription images, voice recordings, transcripts, or health data may be exposed through logs, repositories, or external services.	Obtain microphone permission; minimize or avoid audio retention; use synthetic/de-identified test data; restrict access; exclude secrets/data from Git; review evidence before publication.	Whole team	Open
R-08	Medium	Severe-interaction reminder rules may be bypassed or may incorrectly block unrelated medication.	Implement and test the safety gate per medication; require review/confirmation for affected new reminders; preserve explicit status and audit evidence.	Thành	Open

R-09	Medium	Three-person capacity or uneven workload may delay the five-sprint plan.	Keep a prioritized MVP; assign one owner per item; limit work in progress; review blockers weekly; cross-review and redistribute work at mid-sprint.	Trang	Open
R-10	Medium	Integration may occur too late, causing incompatible NLP, OCR, speech-to-text, UI, service, or data contracts.	Define a common verified-text contract in Sprint 2; deliver thin slices in Sprint 3; integrate continuously; maintain automated smoke tests and tagged baselines.	Whole team	Open
R-11	Medium	The lack of real professional integration may make review behavior unrealistic.	Mark it as a simulation; validate flows with available healthcare feedback; avoid claims of clinical deployment readiness.	Long / Trang	Open

4.3.4 Configuration Management